

Multi-Modal Sensor Prototype Design Documentation

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Report Overview:

This report will overview the details regarding the material specifications of the flex circuit board, the specific circuit configuration and microcontroller architecture, along with potential drawbacks to larger scale implementation and possible solutions to counteract these drawbacks and make the product efficient.

Material Specification and Mechanics:

For the specific board we will be using

- **Base Material:** Polyimide (PI)
- **Substrate Thickness:** 0.1mm
- **Layer Count:** 2-Layer
- **Copper Weight:** 1 oz (or 0.5 oz for maximum flexibility)
- **Stiffeners:** Polyimide stiffeners directly under sensor ICs only.

Compared to a standard FR4 Board in pcb design it can be very difficult to implement into a soft robot due to its substrate having a high young modulus that creates stress concentration in the area it's encapsulated. This problem is inherent especially when the actuators are in activation, the board will be an anchor to the movement of the gripper and hinder its ability to move whilst potentially irritating the wiring of the PCB on the robot.

However, for our flex board; the 0.1mm polyimide substrate is much more suited for a non rigid environment like a soft robots gripper as it doesn't create stress points like a FR4 Board when encapsulated in a silicon like material in our apparatus due it not stretching longitudinally. Keeping the boards substrate at 0.1mm allows the actuator to bend with minimal force and fixes the problem of damaging the board or material it's encapsulated in like an FR4 board.

In our design, we also have to factor in the copper tracing of the FPCB, of course even copper can provide mechanical stiffness to an actuator when bending as it is still a metal. To overcome this, the protocol of communication used was I2C (Inter-Integrated Circuit) this specific protocol will minimize the copper traces used in connection compared to other protocols more over on this in the architecture section. These copper traces will be in between the substrate and cover layers in the middle as this specific neutral point will have the least amount of strain preventing the traces snapping on a bend.

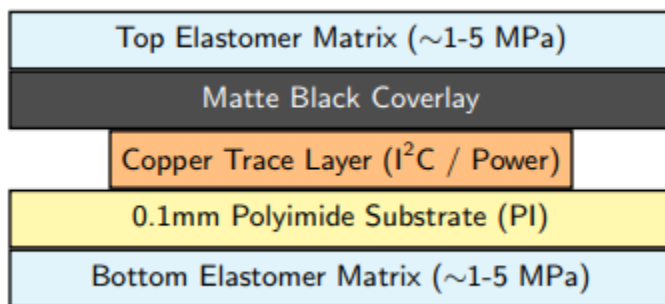


Figure: Layering Diagram of FPCB

Circuit Architecture and Design Logic:

Before getting into the specific circuitry the design relies on 4 main embedded components

Table: Specific Components Used and How They Work

Component	Functions	Data Acquisition	Activation Timing
Master MCU	Bus Master (controls all sensor communications)	Acts as the brain and initiates the initial clocking pulses to the sensors and can request data from specific sensors at all times.	Active at all times to continuously read all sensor data.
MPU-6050	Kinesthetic Tracking	Contains very small silicon masses inside contained by springs inside a capacitor, when gripper moves the sensor reads movement by the inertia of the masses inside the spring and causes the capacitors plates to shift which changes voltage ($V = E \cdot D$) and this change is read after being pushed through an ADC.	100 Hz (polled by master MCU)
BMP280	Tactile Feedback	Has a flexible silicon membrane with an internal resistance, when the gripper touches or holds something the bend on the silicon will alter the resistance and the change is used to measure the grippers force.	100 Hz (polled by master MCU)
LM75	Thermal Feedback	Uses a semiconductor with internal forward biased diodes, if the gripper touches something hot the thermal energy converts to voltage which the diodes read and conducts the voltage drop to ADC.	10 Hz (polled at lower frequency as thermal energy changes much slower then motion which optimizes bus bandwidth compared to 100 Hz)

The primitive components used were capacitors of 0.1 Micro Farads and 4.7 Kilo Ohm pull up resistors. Specifically, the capacitors are used for decoupling which essentially means it filters very high signal noise from the 3.3 Vcc rail to ground preventing a voltage drop which can alter the data read by the controllers with high frequency swinging. For the pull up resistors, they are connected from the supply voltage to the serial data and clocking pins of the sensors, these connections will prevent the controllers from frying with current overload if they communicate with each other the 3.3V high signal and 0V low signal sent will not be across a short circuit but instead through 4.7 Kilo Ohms of resistance.

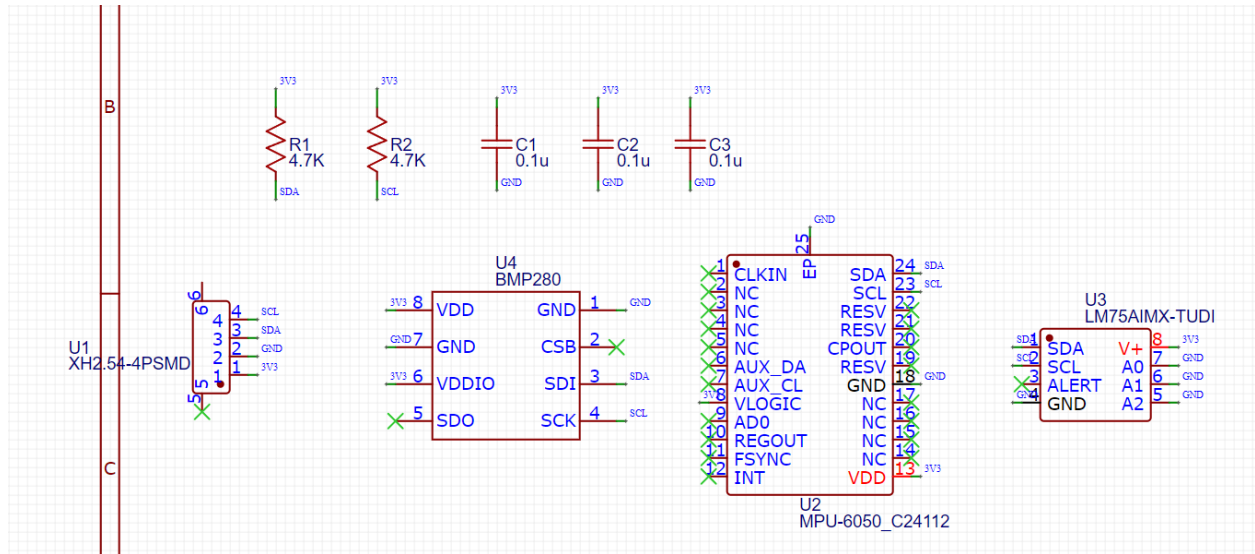


Figure: Multi-Modal Sensor FPCB schematic

Potential Drawbacks and Optimization in the Long Run:

As this is only a prototype design, if we were to scale this FPCB on all fingers of the gripper with more advanced tech there would certainly be some problems to look over, specifically with signal degradation and data management.

Over a long cable along the grippers fingers to the main MCU, the parasitic capacitance inside the wires could potentially drop off sensor signals. We propose a custom multi stage MOSFET amplifying circuit that keeps this signal packed at an optimal frequency to mitigate potential signal degradation. This design is able to maintain a gain of 62 DB with a 13 MHz Bandwidth all whilst drawing under 1 mW of power. The amplifier will be able to filter noise from the parasitic capacitance and boost the signal strength up to 1.61 Vpp swing, which makes sure of optimal data transmission.

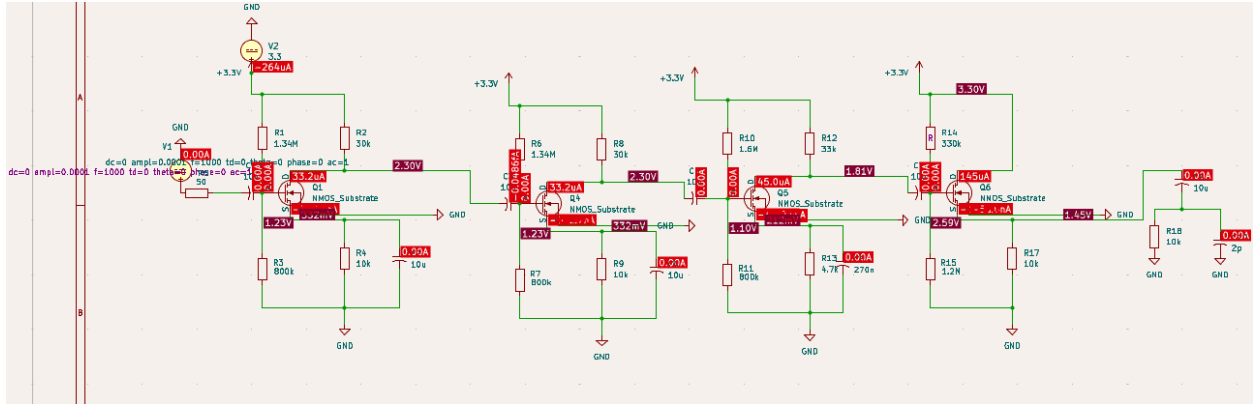


Figure: 4 stage MOSFET amplifier circuit

The current microcontroller is very sufficient for one FPCB connection to orchestrate polling, theoretically it would also work on multiple but polling dozens of sensors with one MCU creates potential delay and processing bottlenecks of sensor reading which makes it very hard for the on board computer of the robot to interpret this from the MCU. Instead, a FPGA (field programmable gate-array) can be mounted on board of the robot where we would wire all the I2C lines of each FPCB to it. This would help astronomically with processing power as an FPGA is very efficient with multi threaded processing tasks and will be able to poll all sensors simultaneously with strict clocking. With an FPGA implementation like this it would be able to packetize the data fast and organized from each sensor to the main on board computer creating virtually no delay; crucial in the application of soft robots.

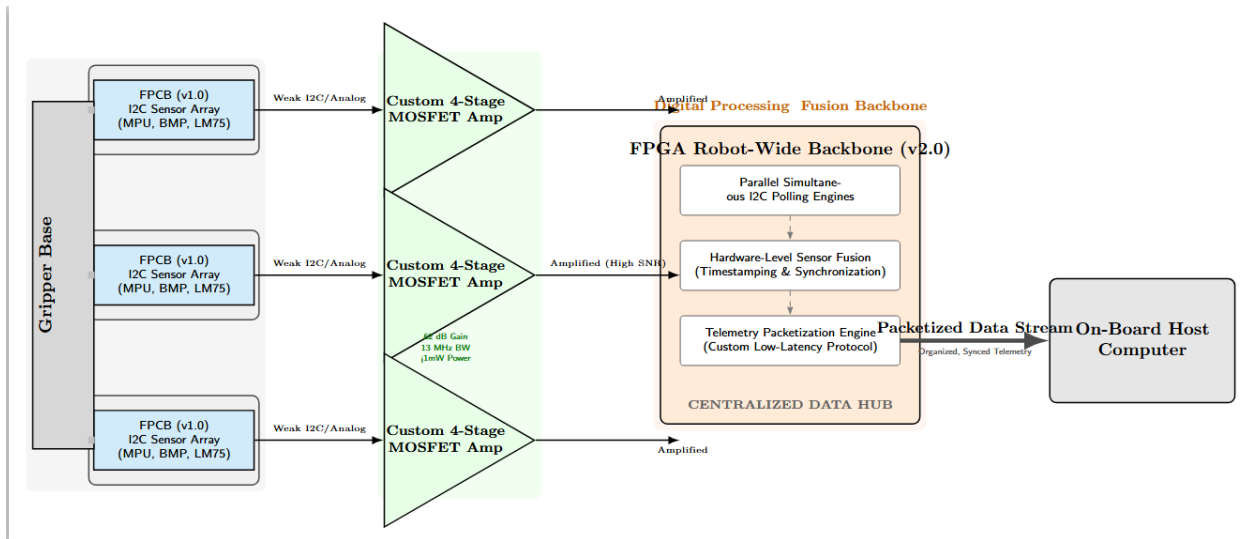


Figure: Block Diagram of Long Run Impl'

Conclusion:

This prototype demonstrates how a 0.1mm Polyimide substrate paired with I2C wiring can create a flexible multi-modal sensor for a soft gripper without compromising itself compared to a rigid PCB implementation. However, as the product scales there are contingencies for failures by creating a MOSFET amplifying configuration with a mounted on-board FPGA to handle signal degradations and data management to cope with real world standards for application in soft robots.